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ChatGPT Prompts for Research and Analysis

If you perform research as part of your work or your studies, you can harness ChatGPT's power to complete that research more quickly. In this chapter, we'll look at how to prompt ChatGPT to summarize articles you might want to read, analyze data you've captured, and suggest topics connected to the topics you're researching. We'll also cover how to prompt ChatGPT to generate research questions for you and how to enlist ChatGPT's help to design experiments.

Ready?

Summarize Articles

If you had infinite time, you could read all the literature on the subject you need to study or to understand. But as your time is limited, you may want to get ChatGPT to summarize one or more articles for you. This can be a great way to get a quick overview of an article — enough, at least, to determine whether you should read the article yourself.

WARNING As of this writing, ChatGPT seems to have a hard time identifying satire, irony, and sarcasm. ChatGPT's summary of an article that includes these elements may make the article sound serious.

If you enable the Web Browsing Model, ChatGPT can summarize articles on the web, provided they are not hidden behind paywalls or otherwise restricted. Give a prompt like this:

PROMPT **Summarize *URL***

ChatGPT reads the article and returns a summary.

Without the Web Browsing Model enabled, you must paste into the Send a Message box the content you want ChatGPT to summarize. Give the "Summarize" prompt, paste the material, and press Enter.

PROMPT **Summarize *pasted_material***

This method works fine, and is usually quicker than having ChatGPT access the material on the web, but it means that the content can be only up to 4096 tokens long (including the "Summarize" keyword), so you cannot summarize longer articles.

Leaving the "Summarize" prompt open like this lets ChatGPT decide on the length of summary to create. Results vary depending on how long the article is and what its contents are, but ChatGPT typically delivers an introductory paragraph, a number of key points, and a concluding paragraph.

If you want ChatGPT to keep the summary to a certain length, give the limit. For example:

PROMPT **Summarize in 50 words *URL***

Alternatively, you might ask for a specific number of key points. For example:

PROMPT **Summarize in three key points *URL***

Analyze Data

ChatGPT can help you to analyze data you have captured. ChatGPT can perform various types of analysis, including these five types:

- **Descriptive analysis.** This simple form of analysis provides a basic understanding of the data by calculating simple metrics, such as the mean, median, mode, and standard deviation.
- **Correlation analysis.** This type of analysis enables you to determine if there's a correlation between two variables. (Spoiler: Correlation does not prove causation.)
- **Regression analysis.** This more sophisticated form of correlation analysis tries to show how one variable is affected by one or more other variables.
- **Time series analysis.** This form of analysis helps you understand trends and seasonality in data collected over an extended period of time.
- **Sentiment analysis.** This form of analysis examines the sentiment expressed in text — for example, in a customer review of a product.

Here are three examples of prompts you could use to ask ChatGPT to analyze data. For each prompt, you would give the data to ChatGPT by pasting it into the Send a Message box after the prompt and (optionally) a line break (press Shift+Enter).

PROMPT **Could you analyze the following data and tell me the mean, the median, the mode, the variance, and the standard deviation?**

ChatGPT spelled out the meaning of the five measures, demonstrated how to work out each of them, and then provided this summary:

So, for this data set:

- *Mean = 57.14*
- *Median = 60*
- *Mode = 80*
- *Variance = 420.33*
- *Standard Deviation = 20.50*

PROMPT **Here's some data from a survey we did that shows the length of customer service calls in seconds and the customer satisfaction rating on a scale of 1 (worst) to 10 (best). Have a look at them and tell me if there's a correlation between the two.**

The data you've provided seems to suggest a positive correlation between the length of customer service calls and the customer satisfaction rating. This is because as the length of the call increases, the customer satisfaction rating also increases.

However, to accurately determine the correlation, you would typically calculate the Pearson correlation coefficient. . . .

ChatGPT explained that it doesn't currently have the capability to calculate mathematical or statistical functions. It suggested using a programming language, such as R or Python, to perform the calculation — and provided a complete Python script for calculating the correlation coefficient.

PROMPT **Hi ChatGPT! I'm giving you ten customer reviews for our new chewable mouse mat. Do a quick sentiment analysis on them, would you?**

ChatGPT analyzed each of the customer reviews, determining whether it expressed positive sentiment, negative sentiment, neutral sentiment, or mixed sentiment. Here are three examples:

1. ***"This product is useless, but it made me laugh."*** *- Mixed sentiment. Negative due to "useless" but also positive because it made the customer laugh.*

4. ***"Fun for 5-year-olds."*** *- Positive sentiment. "Fun" indicates enjoyment or amusement.*

10. ***"This may be chewable, but the flavor is disgusting."*** *- Negative sentiment. "Disgusting" is a strong negative word.*

Reducing text to a value (such as "negative sentiment") feels somewhat grotesque, but greatly simplifies the task of analyzing complex input. ChatGPT does warn that automated tools "may not always interpret sentiment accurately, especially when it comes to sarcasm, irony, or culturally specific expression"; keep this in mind if you use sentiment analysis.

Suggest Connected Topics

If you need to look into topics connected to what you are studying, you can ask ChatGPT for help in identifying them. Here are three examples of prompts you could use:

PROMPT **I'm examining the positive and negative effects of social media usage during the pandemic. Please suggest three connected topics for me to look at.**

ChatGPT suggested Mental Health and Social Media, Social Media as a Source of Information and Misinformation, and Social Media and Physical Activity.

PROMPT **Hey, ChatGPT, I'm studying the effect of daytime napping on academic performance in college students. Could you give me five related topics to consider?**

ChatGPT suggested topics including Sleep Deprivation and Cognitive Function, Impact of Stress on Sleep Quality, and the Role of Circadian Rhythms in Sleep and Academic Performance.

PROMPT **I'm working on the effect that reducing the serving size of fast-food takeout dishes will have on the rat population of major urban areas. Give me 10 connected topics I should think about. Thanks!**

Certainly, your topic is an interesting intersection of public health, urban ecology, and behavioral science. Here are ten related topics that you might consider:

ChatGPT's suggested topics included Urban Waste Management, Rat Behavior and Diet, Impact of Fast-Food Industry on Urban Ecosystems, and Fast-Food Consumption Trends.

Generate Research Questions

ChatGPT can be a valuable resource when you need to generate research questions to probe an area of interest. Instead of you having to cudgel your brains to produce the questions, you can have ChatGPT suggest them in seconds from its vast reading and data banks.

Start by telling ChatGPT the topic in which you're interested and identifying the gaps in current knowledge. Here's an example:

PROMPT **We know that exercise affects appetite. I want to research how the intensity of exercise changes the effect on appetite. For example, some people report being ravenously hungry after intense exercise, whereas other people don't want to eat at all for several hours after intense exercise. Can you generate some research questions for me on this subject?**

ChatGPT returned seven potential research questions, including these three:

Is there a correlation between the intensity of exercise and the types of food (e.g., protein, carbohydrates, fat) that individuals crave post-exercise?

How does intense exercise influence the physiological hunger and satiety hormones (like ghrelin and leptin) compared to moderate or light exercise?

Is there a difference in post-exercise appetite between aerobic exercises (like running) and anaerobic exercises (like weightlifting) of similar intensity?

Here's another example:

PROMPT **I'm investigating the roles of different activities in the acquisition of a foreign language. I want to compare the helpfulness of spoken role-play, in-class group exercises (also spoken), and written exercises. Which activities help students learn most quickly? Please generate eight research questions I could use.**

ChatGPT returned the eight questions I requested, including these three:

How does the complexity level of the target language affect the effectiveness of spoken role-play, in-class group exercises, and written exercises?

How does student engagement and motivation change with each of the three activities (spoken role-play, in-class group exercises, and written exercises) and how does this influence language acquisition speed?

Which of the three activities (spoken role-play, in-class group exercises, or written exercises) contributes more effectively to long-term retention of a foreign language?

Design Experiments

ChatGPT can help you to design experiments that you can perform to further your research. Normally, you'd start off with a research question — for example, a question that your instructor has given you.

Understand the 10 Steps in Designing an Experiment

Designing an experiment typically involves the following 10 steps:

1. Defining the experiment's objective.
2. Developing a hypothesis that predicts the experiment's outcomes.
3. Designing a methodology that suits your research question.
4. Identifying the experiment's independent variables, dependent variables, and controlled variables.
5. Deciding how to collect data for the experiment.
6. Choosing a suitable sample size and appropriate selection techniques.
7. Selecting suitable analysis methods to analyze the data collected.
8. Assessing any ethical considerations involved in the experiment.
9. Identifying any potential sources of error in the experiment.
10. Reporting the results of your experiment.

We'll now work through those 10 steps explained in the previous section. We'll use this research question:

> For overweight and obese adults, does a low-carbohydrate diet improve weight-loss outcomes compared to a standard American diet?

1. Define the Objective

Let's start by defining the objective. You might prompt like this:

PROMPT **I need to design an experiment for the following research question. First, please help me to define the objective of the experiment.**

After the prompts, you'd press Shift+Enter to insert a line break, and then press Ctrl+V to paste in the objective.

ChatGPT returned the objective with some suggestions:

The objective of your experiment, based on your research question, would be:

"To compare the effectiveness of a low-carbohydrate diet versus a standard American diet on weight-loss outcomes in overweight and obese adults."

To further refine your objectives, you may want to define what specific weight-loss outcomes you're interested in. For example:

- *Total weight loss after a certain period (e.g., 3 months, 6 months).*
- *Changes in body mass index (BMI).*
- *Changes in body fat percentage.*
- *Long-term maintenance of weight loss (e.g., after 1 year).*

2. Develop a Hypothesis

Next, we need to develop a hypothesis that predicts the experiment's outcomes. We can use a prompt such as this:

PROMPT **Please suggest a hypothesis based on that objective.**

ChatGPT suggested the following hypothesis:

"Overweight and obese adults following a low-carbohydrate diet will experience greater weight loss outcomes compared to those following a standard American diet over a defined period."

ChatGPT also suggested using a specific measure for assessing the weight-loss outcomes, such as total weight loss, decrease in BMI, or decrease in body-fat percentage. So I prompted:

PROMPT **That hypothesis is good, but please add "total weight loss" as the measure.**

ChatGPT made that change.

3. Design a Methodology for the Experiment

The next step is to design a methodology that suits the research question. We can prompt ChatGPT like this:

PROMPT **Design a methodology for the experiment.**

ChatGPT explained that we need to design a controlled experiment — one that has a control group that will enable us to tell whether our intervention has worked. We will recruit enough overweight and obese adults to make a sample, explaining the experiment to them and making sure they understand the purpose and the risks. We then assign the subjects randomly to the intervention group and the control group, perform the intervention for our determined period, and measure the results.

4. Identify the Independent, Dependent, and Controlled Variables

Next, we need to identify the experiment's independent variables, dependent variables, and controlled variables. ChatGPT can spell these out:

PROMPT **What are the experiment's independent variables, dependent variables, and controlled variables?**

ChatGPT explained that the independent variable is the one we manipulate: the participants' type of diet. The dependent variable is the variable we measure: the participants' weight. The controlled variables are variables we keep the same for all participants, such as their total caloric intake, activity levels, and frequency of weight measurement.

5. Decide How to Collect Data

The next step is to decide how to collect data for the experiment. I prompted ChatGPT like this:

PROMPT **How should I collect data for the experiment?**

ChatGPT explained that the experiment needs at least the weight, but other measurements may be useful, too. We may want to measure height so we can calculate BMI, measure body fat percentage, or measure other health indicators.

ChatGPT walked me through initial data collection, ongoing data collection at regular intervals during the study period, and final data collection at the end of the study.

6. Choose the Sample Size and Selection Techniques

Next, you need to choose a suitable sample size and appropriate selection techniques for the study. I prompted ChatGPT like this:

PROMPT **What sample size and selection techniques should I use for the experiment?**

ChatGPT explained that the larger the sample size, the more closely it will represent the population and the greater its statistical significance will be — but that I also need to work within my resources. ChatGPT mentioned that I could run a power analysis to calculate the appropriate sample size.

Similarly, ChatGPT pointed out that, while random selection of participants would give a truer representation of the population, convenience sampling (in lay terms, taking whatever participants you can get) is likely to be more practical.

7. Select Suitable Analysis Methods

The seventh step is to select suitable analysis methods to analyze the data collected. I prompted ChatGPT for advice like this:

PROMPT **What analysis methods should I use to analyze the data collected in the experiment?**

ChatGPT recommended calculating descriptive statistics for the intervention group and the control group, such as calculating the mean and the medium to measure the central tendency of the weight loss and calculating the range and the standard deviation to measure its dispersion. Next, ChatGPT suggested moving on to inferential statistics and using a t-test if the data meets the assumptions for parametric tests, or using a non-parametric test, such as the Mann-Whitney U test, if it does not.

If I collected data on confounding variables, I could use multivariate regression or analysis of covariance (ANCOVA) to control for them.

Whatever the other results, I should calculate the effect size to measure the size of the difference between the control group and the intervention group.

8. Assess Any Ethical Considerations

Next, I should assess any ethical considerations involved in the experiment. To this end, I prompted ChatGPT thus:

PROMPT **Please assess any ethical considerations involved in the experiment.**

ChatGPT produced a comprehensive list, starting with Informed Consent and Confidentiality and ending with Fair Participant Selection, Ethics Committee Approval, and Scientific Integrity.

9. Identify Potential Sources of Error

Next, I need to identify potential sources of error in the experiment:

PROMPT **Identify any potential sources of error in the experiment.**

ChatGPT returned a list of potential sources of error, such as dietary adherence, physical activity levels, measurement errors, and confounding variables (such as sleep patterns, stress levels, or genetic factors).

10. Report the Results

All that remains is to report the results of the experiment. ChatGPT is happy to provide advice on how to do this. I prompted:

PROMPT **How can I report the results of the experiment?**

ChatGPT recommended presenting the findings, methods, and results in the form of a scientific research paper, which would have these sections:

- Title
- Abstract
- Introduction
- Methods
- Results
- Discussion
- Conclusions
- References